

Probabilistic Spatiotemporal Air Quality Forecasting via Physics-Guided Attention and Variational Inference

July 30, 2026

Introduction

The evolution of society has brought about several irreversible developments, one of the most notable being industrialization. These advancements have had a profound impact on nearly every aspect of life; however, one of the most critical consequences concerns human health and overall well-being, among which air quality has emerged as a major issue. This prominence stems from the excessive emissions generated by industrialization and transportation activities, both of which have become inseparable parts of modern society. Harmful pollutants such as PM2.5, SO, CO, and CO released from these activities significantly degrade air quality. Due to their ability to disperse and circulate easily in the atmosphere, these pollutants expose a large population to hazardous conditions, particularly in densely populated urban areas where both human activity and emission sources are highly concentrated.

A wide range of efforts has been made to mitigate these impacts, ranging from data collection and analysis to the development of mitigation strategies. Among these, air quality analysis plays a critical role by enabling more accurate, evidence-based, and effective decision-making. In particular, air quality forecast with high accuracy is considered an effective way to mitigate detrimental impact of pollutant exposure [6]. Consequently, air quality forecasting has become an important research area in environmental science. Traditional forecasting methods primarily rely on historical observations to predict future values over time. Nevertheless, because air quality exhibits strong spatial dependency, incorporating spatial information can effectively complement temporal patterns. This integration provides richer contextual information, analogous to understanding the surrounding environment in addition to observing how a phenomenon evolves over time. As a result, spatial-temporal forecasting models can better capture the underlying dynamics, leading to more accurate, representative, and realistic predictions.

Related Studies

Machine learning (ML) models have been widely employed due to their inherent ability to learn the mapping between inputs and outputs by optimizing model parameters to minimize prediction

loss. This capability enables ML models to adapt to the underlying characteristics of a dataset, making them effective tools for investigating historical patterns and predicting future values. Several popular ML methods applied to air quality forecasting belong to the boosting family, including Gradient Boosting, XGBoost [15], LightGBM [11], and CatBoost [9].

Deep learning represents a more advanced and sophisticated class of machine learning models. Although often regarded as "black-box" models due to their limited interpretability, their remarkable ability to capture complex, nonlinear relationships between input features and target variables has made them a dominant approach in forecasting research. Long Short-Term Memory (LSTM) networks are particularly well known for their capability to preserve long-term sequential dependencies, making them a common choice for temporal modeling [13]. A notable extension of LSTM is the Bidirectional Long Short-Term Memory (Bi-LSTM), which processes sequential information in both forward and backward directions to capture richer temporal representations [?]. To improve computational efficiency while maintaining comparable performance, the Gated Recurrent Unit (GRU) has also been widely adopted as a more compact recurrent architecture [4].

Transformer are a powerful candidate due to full data access or selective data access by its different variants, contrasting limited reach and vanishing gradient issues in other deep learning backbones. Hu et al. [3] adopted a Transformer encoder-decoder as the primary temporal forecasting backbone, while spatial dependencies among neighboring grids were modeled through a cross-optimal interpolation layer, where distance-aware learnable weights were computed using a Softmax-like formulation.

To further leverage the complementary strengths of different learning algorithms, ensemble methods have also been extensively explored. By combining multiple predictive models, ensemble approaches can often achieve improved robustness and forecasting accuracy compared with individual models [10, 8].

Hybrid models appear to be a feasible strength due to the diverse roles accompanied by contributing components and thus allow for multi-purpose models. Deep learning seems to join this very much. For instance, Milli et al., combined Prophet to capture long term evolution while afterward LSTM to learn non linear dependencies. Transformers and deep learning models are good combo. Dong et al., [2] utilized empirical mode decomposition (EMD) to simplified data into multiscale fracture, meanwhile self attention learn data relationship and Bi-LSTM for forecast output. Similar approaches have been developed by Liu et al., [7]. Another expressive model is developed by Wei et al., [16] where CNN, LSTM and Attention is stacked in hierarchical architecture and further enhanced by hyperparameter optimization by Whale optimization algorithm (WOA). Another interesting combination by Inam et al., [5] which utilize a simpler form of deep learning is the combination of Polynomial Regression (PR) and Fully Connected Neural Networks (FCNN), which has showed surpassed performance compared to hybrid model of Random Forest (RF) and deep learning and relative better performance compared to hybrid deep learning with LSTM families going regression and boosting families.

Transformers is also a potential candidate for model construction. Zhang and Awang [19] adopted the Inverted Transformer architecture to overcome the limitation of the vanilla Transformer, which embeds all features at a single time step into one token. Instead, the Inverted Transformer treats each feature as an independent token across the entire time span, thereby emphasizing temporal dependencies throughout the sequence rather than inter-feature correlations at individual time

steps.

Methodology

Graph Construction and Physics-Aware Spatial Attention

Let i denote a neighboring (source) station and j denote the target station. Throughout this section, all pairwise quantities are indexed according to the ordered pair (i, j) , representing the potential transport of pollutants from station i to station j . Since multiple neighboring stations may influence the same target station, multiple values of i may correspond to the same target station j .

Station Graph Construction

Each monitoring station has a fixed geographical coordinate consisting of latitude and longitude. To accurately estimate the distance between stations over a large geographical region, the Haversine formula is adopted instead of Euclidean or Manhattan distance. Unlike planar distance measures, the Haversine distance accounts for the Earth’s curvature and therefore provides a more accurate estimate of great-circle distance between two stations.

The pairwise distance between stations i and j is computed as

$$a_{ij} = \sin^2 \left(\frac{\Delta\phi_{ij}}{2} \right) + \cos(\phi_i) \cos(\phi_j) \sin^2 \left(\frac{\Delta\lambda_{ij}}{2} \right), \quad (1)$$

$$d_{ij} = 2R_{\oplus} \arcsin \left(\sqrt{a_{ij}} \right), \quad (2)$$

where $R_{\oplus} = 6371$ km is the Earth’s radius, ϕ and λ denote latitude and longitude in radians, respectively, and

$$\Delta\phi_{ij} = \phi_i - \phi_j, \quad \Delta\lambda_{ij} = \lambda_i - \lambda_j.$$

For numerical stability, the implementation clips a_{ij} to the interval $[0, 1]$ before applying the square root and inverse sine functions.

Although the pairwise distance describes the spatial separation between two stations, it does not indicate the direction of pollutant transport. Therefore, the geographic bearing from neighboring station i to target station j is also computed. The bearing represents the clockwise angle measured from True North to the line connecting station i to station j , and is expressed in radians as

$$\beta_{i \rightarrow j} = \text{atan2}(\sin(\Delta\lambda_{ji}) \cos(\phi_j), \cos(\phi_i) \sin(\phi_j) - \sin(\phi_i) \cos(\phi_j) \cos(\Delta\lambda_{ji})), \quad (3)$$

where

$$\Delta\lambda_{ji} = \lambda_j - \lambda_i.$$

Since station locations remain fixed throughout training, both the distance matrix and the bearing matrix are precomputed once and stored as static buffers.

Neighbor Graph

PM_{2.5} transport is generally localized over a limited spatial range. Therefore, only stations located within a predefined distance threshold are considered neighbors. In this study, the neighborhood radius is fixed at 300 km, forming a static graph throughout training.

The neighborhood of target station j is defined as

$$\mathcal{N}(j) = \{i \mid d_{ij} \leq 300 \text{ km}\}. \quad (4)$$

Consequently, only neighboring stations within a 300-km radius centered at the target station are considered during spatial attention, while all remaining station pairs are masked.

Scaled Dot-Product Spatial Attention

Unlike the original Transformer, where each token represents a position in a sequence, the proposed model treats each monitoring station as one spatial token. At each time step, the hidden representation of every station participates in the attention mechanism.

Given the hidden representation h_i of station i , three linear projections are first applied to obtain the Query, Key, and Value vectors,

$$q_i = W_Q h_i, \quad k_i = W_K h_i, \quad v_i = W_V h_i, \quad (5)$$

where W_Q , W_K , and W_V are learnable weight matrices.

The content similarity between neighboring station i and target station j is then computed using scaled dot-product attention,

$$s_{ij}^{\text{content}} = \frac{q_i^\top k_j}{\sqrt{d_k}}, \quad (6)$$

where d_k denotes the dimensionality of the Query and Key vectors.

The scaling factor prevents excessively large dot-product values, thereby improving numerical stability before the SoftMax normalization.

Distance Bias

Besides the learned content similarity, a physics-inspired distance bias is introduced to encourage stronger interactions between geographically closer stations. The distance bias is defined as

$$\text{dist}_{ij} = \exp\left(-\frac{d_{ij}}{\sigma_d}\right), \quad (7)$$

where $\sigma_d = 200$ km controls the decay rate. Consequently, nearby stations receive larger bias values, whereas the contribution gradually decreases with increasing geographical distance.

Terrain Bias

Topographical differences also influence pollutant transport. Stations separated by large elevation differences are generally less likely to exchange pollutants due to terrain barriers. Therefore, an elevation-based terrain bias is defined as

$$\text{terrain}_{ij} = \exp\left(-\frac{|e_i - e_j|}{\sigma_h}\right), \quad (8)$$

where e_i and e_j denote the elevations of stations i and j , respectively, and $\sigma_h = 1200$ controls the attenuation caused by terrain differences.

As the elevation difference increases, the terrain bias decreases, reducing the likelihood of pollutant transport between the two stations.

Wind Alignment

Wind direction plays a major role in the transport of atmospheric pollutants. A neighboring station is expected to have a stronger influence when it lies approximately in the direction of the prevailing wind at the source station, while stations located upwind are much less likely to contribute to pollutant transport.

For every ordered station pair (i, j) , the angular difference between the instantaneous wind direction at neighboring station i and the geographic bearing from station i to station j is computed as

$$\theta_{ij}(t) = \psi_i(t) - \beta_{i \rightarrow j}, \quad (9)$$

where $\psi_i(t)$ denotes the wind direction at station i .

The wind-alignment score is then defined as

$$\text{wind}_{ij}(t) = \max(\cos \theta_{ij}(t), 0). \quad (10)$$

The resulting score satisfies

$$0 \leq \text{wind}_{ij}(t) \leq 1.$$

When

$$|\theta_{ij}| < 90^\circ,$$

station j lies approximately in the downwind direction of station i , resulting in a positive alignment score. The alignment reaches its maximum value of one when the wind direction exactly matches the bearing from station i to station j .

Conversely, when

$$|\theta_{ij}| \geq 90^\circ,$$

the cosine becomes non-positive and is clamped to zero. Consequently, the wind-alignment term provides no additional evidence that pollutants are transported from station i toward station j . This does not remove the station pair from the attention mechanism; rather, it only removes the wind-related contribution to the final attention score.

Advection Distance

In practice, PM_{2.5} transport does not necessarily follow the static shortest distance between neighboring station i and target station j . Instead, pollutant movement is influenced by the wind field, where both wind speed and direction determine the displacement of the transported air parcel.

Therefore, rather than measuring the interaction strength using only the stationary distance d_{ij} , we estimate the effective distance after considering the movement of the air parcel during one timestep.

Let A denote the location of the neighboring station i , and B denote the location of the target station j . After one timestep, the air parcel originating from station i is assumed to move from A to A' according to the local wind condition. The objective is then to estimate the distance between the advected air parcel position A' and the target station B , denoted as d_{ij}^{eff} .

The displacement of the air parcel is calculated from the wind speed as

$$\Delta_i(t) = u_i(t) \times 3.6 \times \Delta t, \quad (11)$$

where $u_i(t)$ is the wind speed at station i in m/s, 3.6 is the conversion factor from m/s to km/h, and Δt represents the dataset timestep in hours. In this study, $\Delta t = 3$ hours.

The original distance between station i and station j is obtained from the Haversine distance described previously:

$$AB = d_{ij}. \quad (12)$$

Furthermore, the angle between the wind direction at station i and the direction from station i to station j is defined as

$$\theta_{ij}(t) = \psi_i(t) - \beta_{i \rightarrow j}, \quad (13)$$

where $\psi_i(t)$ denotes the instantaneous wind direction at station i , and $\beta_{i \rightarrow j}$ is the geographic bearing from station i to station j .

Considering the triangle formed by A , A' , and B , the effective distance $A'B$ can be derived using the law of cosines:

$$d_{ij}^{\text{eff}}(t) = \sqrt{d_{ij}^2 + \Delta_i(t)^2 - 2d_{ij}\Delta_i(t)\cos(\theta_{ij}(t))}. \quad (14)$$

This formulation allows the spatial relationship between two stations to dynamically change according to the current wind condition. When the wind blows directly from station i toward station j , $\theta_{ij} = 0$, and the effective distance becomes

$$d_{ij}^{\text{eff}} = |d_{ij} - \Delta_i|,$$

indicating that the transported pollutant parcel moves closer to station j . Conversely, when the

wind blows away from station j , the effective distance increases, reducing the likelihood of pollutant influence.

Finally, the advected distance is converted into an attention bias using an exponential decay function:

$$\text{adv}_{ij}(t) = \exp\left(-\frac{d_{ij}^{\text{eff}}(t)}{\sigma_d}\right), \quad (15)$$

where σ_d controls the spatial decay rate.

The advected distance formulation provides a more physically meaningful representation of spatial dependency compared with conventional distance-based graph construction methods. Existing spatio-temporal PM_{2.5} forecasting approaches typically construct spatial relationships using static geographical distance, where the influence between two stations is assumed to remain unchanged regardless of the prevailing atmospheric conditions. Although distance-based graphs effectively capture local spatial correlation, they do not explicitly consider the dynamic movement of pollutants caused by wind-driven transport.

In contrast, the proposed advected distance introduces a dynamic spatial relationship by estimating the relative position of an air parcel after being transported by the current wind field. Therefore, two stations that are geographically distant may still exhibit strong interactions when the wind direction and speed support pollutant transport, while nearby stations may have reduced influence when the wind moves pollutants away from the target location.

By incorporating wind-driven displacement into the spatial attention mechanism, the model can move beyond purely distance-based assumptions and better capture the directional and dynamic nature of atmospheric pollutant propagation. This provides a more realistic approximation of physical transport processes while maintaining the flexibility of data-driven attention mechanisms.

Combined Attention Score

After computing the individual physical bias terms, they are integrated with the conventional Scaled Dot-Product Attention score to produce the final attention score before normalization.

The content similarity between neighboring station i and target station j is first computed using Scaled Dot-Product Attention:

$$s_{ij}^{\text{content}}(t) = \frac{\mathbf{q}_i(t)^\top \mathbf{k}_j(t)}{\sqrt{d_k}}, \quad (16)$$

where $\mathbf{q}_i$ and $\mathbf{k}_j$ denote the query and key vectors, respectively, and d_k is the dimensionality of the attention space.

Rather than relying solely on this learned similarity, the proposed model augments the attention score with four physics-guided bias terms introduced in the previous subsections:

- distance bias,
- terrain bias,
- wind-alignment bias,
- advected-distance bias.

The combined attention score is therefore expressed as

$$s_{ij}(t) = s_{ij}^{\text{content}}(t) + w_{\text{dist}} b_{ij}^{\text{dist}} + w_{\text{wind}} b_{ij}^{\text{wind}}(t) + w_{\text{terrain}} b_{ij}^{\text{terrain}} + w_{\text{adv}} b_{ij}^{\text{adv}}(t), \quad (17)$$

where each w is a learnable scalar parameter that controls the contribution of its corresponding physical bias.

Unlike manually assigning fixed importance to each physical factor, these coefficients are optimized jointly with the network parameters during training. Consequently, the model can automatically determine the relative importance of geographical distance, terrain difference, wind alignment, and advected distance for the forecasting task. If a particular physical prior contributes little to prediction performance, its corresponding weight can naturally decrease during optimization.

This combined score is subsequently used as the input to the SoftMax function to obtain the normalized attention weights. In the following subsection, an additional source concentration bias is introduced to further account for the amount of $\text{PM}_{2.5}$ available for transport from each neighboring station.

Source Concentration Bias

While the advected distance describes the likelihood that pollutants can be transported from a neighboring station to a target station, it does not account for the amount of pollutant available at the source. In practice, atmospheric transport depends not only on the transport pathway but also on the pollutant concentration carried by the moving air parcel. Consequently, a station with very low $\text{PM}_{2.5}$ concentration should contribute less to the target station than a highly polluted station, even if both exhibit similar wind conditions and geographical proximity.

To incorporate this physical intuition, the proposed model introduces a source concentration bias that combines the advected distance with the instantaneous $\text{PM}_{2.5}$ concentration measured at the source station.

Let $P_i(t)$ denote the $\text{PM}_{2.5}$ concentration at neighboring station i at time t . Since $\text{PM}_{2.5}$ concentration is non-negative, the implementation first ensures numerical stability by clipping negative values and adding a small constant

$$P_i^+(t) = \max(P_i(t), 0) + \varepsilon, \quad (18)$$

where $\varepsilon = 10^{-3}$ prevents undefined values when computing the logarithm.

The source concentration bias is then defined as

$$b_{ij}^{\text{mass}}(t) = -\frac{d_{ij}^{\text{eff}}(t)}{\sigma_d} + \log(P_i^+(t)), \quad (19)$$

where $d_{ij}^{\text{eff}}(t)$ is the effective distance derived in the previous subsection and σ_d is the same spatial decay coefficient used in the distance and advected-distance biases.

Using the logarithmic identity

$$\log(a) + \log(b) = \log(ab),$$

the above expression can be interpreted as

$$b_{ij}^{\text{mass}}(t) = \log\left(P_i^+(t) \exp\left(-\frac{d_{ij}^{\text{eff}}(t)}{\sigma_d}\right)\right). \quad (20)$$

Therefore, the bias is proportional to the product of two physically meaningful quantities: the pollutant concentration available at the source station and the transport efficiency determined by the wind-driven effective distance. A neighboring station with a high $\text{PM}_{2.5}$ concentration and a favorable transport pathway naturally receives a larger attention bias than a clean station or one located along an unfavorable transport path.

The logarithmic transformation serves two purposes. First, it compresses the dynamic range of $\text{PM}_{2.5}$ measurements, preventing the concentration term from overwhelming the other attention components. Second, it converts the multiplicative relationship between source concentration and transport efficiency into an additive bias that can be directly incorporated into the attention score.

Finally, the source concentration bias is added to the overall attention score as

$$s_{ij}(t) = s_{ij}(t) + w_{\text{mass}} b_{ij}^{\text{mass}}(t), \quad (21)$$

where w_{mass} is a learnable scalar parameter that automatically determines the contribution of the source concentration bias during training. If this information provides little predictive value, the optimization process can naturally reduce w_{mass} toward zero, allowing the model to rely more heavily on the remaining attention components.

Advective PM_{2.5} Transport Feature

Although the proposed spatial attention mechanism incorporates several physically-inspired bias terms, the attention operation itself is performed on latent hidden representations rather than directly on the raw PM_{2.5} observations. Since each hidden state jointly encodes historical PM_{2.5} concentrations together with multiple meteorological variables, the contribution of neighboring PM_{2.5} sources is learned implicitly during feature extraction. However, from the perspective of atmospheric transport, the amount of pollutant available at neighboring stations is itself an important physical indicator of the pollutant likely to reach the target station.

To explicitly provide this information to the temporal encoder, an additional feature termed the *Advective PM_{2.5} Transport Feature* is introduced. Unlike the spatial attention module, which exchanges learned hidden representations after encoding, this feature is computed directly from the raw PM_{2.5} observations and is concatenated with the original input features before entering the GRU encoder. Consequently, the GRU receives an explicit estimate of the PM_{2.5} transported from surrounding stations while still learning higher-level latent representations through the attention mechanism.

The computation follows the same physical assumptions introduced in the Advective Spatial Attention module. First, the effective transport distance between neighboring station i and target station j is computed using the wind-driven displacement model,

$$d_{ij}^{\text{eff}}(t) = \sqrt{d_{ij}^2 + \Delta_i(t)^2 - 2d_{ij}\Delta_i(t)\cos\theta_{ij}(t)}, \quad (22)$$

where d_{ij} is the Haversine distance, $\Delta_i(t)$ is the wind-driven displacement over one timestep, and $\theta_{ij}(t)$ is the angle between the wind direction at station i and the bearing from station i to station j .

Instead of directly using the effective distance, the transport efficiency is modeled using two physically motivated components. The first component describes attenuation during transport,

$$\omega_{ij}^{\text{path}}(t) = \exp\left(-\frac{d_{ij}^{\text{eff}}(t)}{L_{\text{decay}}}\right), \quad (23)$$

where L_{decay} is a learnable decay length controlling how rapidly pollutant influence decreases with transport distance.

The second component models atmospheric dilution caused by stronger winds,

$$\omega_i^{\text{dilution}}(t) = \frac{1}{1 + \kappa u_i(t)}, \quad (24)$$

where $u_i(t)$ denotes the wind speed at the neighboring station and κ is a learnable parameter controlling the dilution rate. Higher wind speeds increase atmospheric mixing and dispersion, thereby reducing the pollutant concentration arriving at the target station.

The overall transport coefficient is obtained by multiplying the two components,

$$\omega_{ij}(t) = \omega_{ij}^{\text{path}}(t) \omega_i^{\text{dilution}}(t), \quad (25)$$

while stations outside the predefined neighborhood radius are assigned zero transport weight.

Finally, the transported PM_{2.5} feature for target station j is computed as the normalized weighted average of neighboring PM_{2.5} concentrations,

$$T_j(t) = \frac{\sum_{i \in \mathcal{N}(j)} \omega_{ij}(t) P_i(t)}{\sum_{i \in \mathcal{N}(j)} \omega_{ij}(t)}, \quad (26)$$

where $P_i(t)$ denotes the observed PM_{2.5} concentration at neighboring station i . The normalization prevents stations with a larger number of neighbors from producing disproportionately large transport estimates, making the feature comparable across the monitoring network.

Compared with the spatial attention mechanism, which determines how encoded hidden representations are exchanged among stations, the proposed Advective PM_{2.5} Transport Feature provides the encoder with an explicit, physics-guided estimate of incoming pollutant concentration before feature extraction begins. Consequently, the model benefits from both an explicit transport-related input feature and a learned spatial interaction mechanism, allowing the influence of neighboring PM_{2.5} sources to be modeled at two complementary stages of the forecasting pipeline.

Experimental Design

Dataset Description

We evaluate the proposed model on the **KnowAir** dataset [1], a large-scale benchmark for air quality forecasting introduced by PM2.5-GNN [14]. The dataset has subsequently been adopted by numerous state-of-the-art studies, including Air-DualODE [12], AirDDE [17], and EGCFC [18], enabling fair comparisons across different forecasting models.

KnowAir covers a broad region of 184 nodes in China spanning approximately 103°E–122°E and 28°N–42°N, encompassing several heavily polluted urban clusters. Air quality observations are collected from monitoring stations provided by the Ministry of Ecology and Environment (MEE), while meteorological information is obtained from the ERA5 reanalysis dataset released by the European Centre for Medium-Range Weather Forecasts (ECMWF). The dataset records measurements every three hours from January 1, 2015 to December 31, 2018 (GMT).

Following previous studies, PM_{2.5} concentration is the prediction target. Each sample combines historical pollutant observations with multiple meteorological variables describing atmospheric conditions that influence pollutant transport, accumulation, and dispersion. The meteorological variables included in KnowAir are summarized in Table 1.

Table 1: Meteorological variables included in the KnowAir dataset.

Category	Variables
Wind	100m U-component, 100m V-component, 950 hPa U-component, 950 hPa V-component
Temperature	2m temperature, 2m dewpoint temperature, 925 hPa temperature, 950 hPa temperature
Humidity	Relative humidity (950 hPa, 975 hPa), Specific humidity (950 hPa)
Atmospheric Conditions	Surface pressure, Boundary layer height, Vertical velocity (950 hPa), Vorticity (950 hPa), K-index
Precipitation	Total precipitation

The complete dataset spans four years, from January 2015 to December 2018. Following the official benchmark settings, three experimental protocols are provided for evaluating forecasting performance under different temporal scenarios. Protocol 1 uses continuous yearly splits, while Protocols 2 and 3 focus on seasonal and short-term forecasting, respectively. The data partitions are summarized in Table 2.

Table 2: Official data splits of the KnowAir dataset.

Protocol	Training	Validation	Testing	Frequency (hours)
1	2015/01 – 2016/12	2017/01 – 2017/12	2018/01 – 2018/12	3
2	2015/11 – 2016/02	2016/11 – 2017/02	2017/11 – 2018/02	3
3	2016/09 – 2016/11	2016/12	2017/01	3

Evaluation Metrics

To quantitatively evaluate the prediction performance of each model, two widely used regression error metrics are employed: the Root Mean Square Error (RMSE) and the Mean Absolute Error (MAE). The formulations of these metrics are given as follows:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (27)$$

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (28)$$

where N denotes the total number of observations, y_i is the ground-truth PM_{2.5} concentration of the i -th sample, and $\hat{y}_i$ is the corresponding predicted value.

Configuration and Benchmark Models

The experimental configuration used throughout this study is summarized in Table 3. To ensure a fair and objective comparison among different models while mitigating the effect of random weight initialization and stochastic optimization, each experiment was independently repeated five times using different random seeds. The reported performance is the average of these five runs.

Table 3: Experimental configuration.

Hyperparameter	Value
Batch size	32
Epochs	50
Learning rate	5×10^{-4}
Weight decay	5×10^{-4}
History length	24
Prediction length	24
Early stopping patience	10
Repeated runs	5

The dataset has a temporal granularity of three hours. By taking the historical timestep of 24, the model forecasts the next 24 steps, meaning that the past three days of records are used to estimate air quality conditions over the following three days. This setup is considered sufficient to provide adequate historical information while also maintaining a practical and realistic forecasting horizon for various application scenarios. This configuration is adopted as the standard benchmark setting among the evaluated models.

All experiments were conducted on a workstation equipped with an NVIDIA Tesla T4 GPU with 16 GB GDDR6 memory. The models were implemented using Python 3.13 and PyTorch 2.6.0 with CUDA 12.4 and cuDNN 9.1 acceleration. All experiments were performed using a single GPU.

Result and Discussion

Ablation Studies

An ablation study provides a more comprehensive understanding of the proposed model by systematically evaluating the impact of individual components through their inclusion or removal. This analysis reveals the contribution of each component to the overall model performance, highlighting its effectiveness and importance within the complete architecture. Furthermore, the ablation study also investigates the effects of different hyperparameter settings, such as the lengths of the historical input sequence and the forecasting horizon. Although longer historical sequences are generally expected to provide more informative temporal patterns and improve prediction accuracy, this assumption does not always hold in practice. Excessively long input sequences may introduce redundant or noisy information, potentially degrading model performance. Therefore, evaluating different parameter configurations not only provides deeper insights into the model's behavior but also helps identify an optimal experimental setup that balances predictive accuracy and computational efficiency. In general, three categories of ablation studies are conducted: (1) the effects of different input and output sequence lengths, (2) comparisons of alternative temporal modeling architectures, and (3) component ablation through the removal of individual modules. These experiments provide a comprehensive evaluation of the proposed model by examining the influence of sequence configuration, temporal feature extraction, and each architectural component on the overall forecasting performance.

Input/Output sequence length

Input length and Output length are considered important configurations as it determines the amount of data or the complexity of the outcomes when output length is large.

Temporal Modeling Architectures

The core backbone for sequential processing is Gated Recurrent Unit (GRU), which has been intentionally designed for forecasting tasks at limited expense. Still it is worth-noting that there are many other models specifically appended for this task as well, with varying competent in accuracy and resource consumption. Thus experiment on those model is worth. The 3 popular model to be tested in place of GRU are MLP, LSTM, and Temporal Convolutional Network (TCN.)

Conclusion

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